

SAVYA KHOSLA

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RESEARCH INTERESTS

Representation Learning: Creating efficient and scalable visual representations that can substitute traditional patch-based encoding

Long-Form Video Understanding: Enabling fine-grained retrieval and understanding from episodic memory

Multimodal Learning: Developing unified systems that understand modalities like vision, language, audio, and even bio-signals

EDUCATION

University of Illinois Urbana-Champaign

Ph.D. in Computer Science

Urbana, IL

Aug 2024 - Present

- Advisors: Prof. Derek Hoiem and Prof. Alex Schwing

University of Illinois Urbana-Champaign

M.S. in Computer Science (CGPA: 4.0 / 4.0)

Urbana, IL

Aug 2022 - May 2024

- Advisor: Prof. Derek Hoiem
- Thesis: [Dynamic Multimodal Learning](#)

Delhi Technological University

B.Tech. in Computer Engineering (CGPA: 9.40 / 10.0)

New Delhi, DL

Aug 2017 - July 2021

- Advisor: Prof. Anil Singh Parihar

RESEARCH EXPERIENCE

Allen Institute for AI

Research Intern

Seattle, WA

May 2026 - Dec 2026

- Designing content-adaptive visual tokenization for scalable vision-language models, reducing visual token count by 8x while retaining performance on fine-grained tasks like dense image captioning and spatial grounding through pointing

Meta Reality Labs

Research Scientist Intern

Burlingame, CA

May 2025 - Oct 2025

- Built EMG-JEPA, a self-supervised model for learning robust representations from continuous EMG time-series data

University of Illinois Urbana-Champaign

Research Fellow

Urbana, IL

Aug 2024 - May 2025

- Developed REN [[NeurIPS 2025](#)], a region-based image encoder that represents each semantically meaningful region with a token
- Achieved 25x reduction in token count compared to patch encoders while outperforming them on segmentation and retrieval tasks

Adobe Research

Research Scientist Intern

San Jose, CA

May 2024 - Aug 2024

- Proposed MAGNET [[ACL 2025](#)], a unified framework that can enhance any off-the-shelf LLM with representation learning and infilling capabilities, while retaining its core generative functions

University of Illinois Urbana-Champaign

Research Assistant (Advisor: Prof. Derek Hoiem)

Urbana, IL

Jan 2024 - May 2024

- Developed RELOCATE [[CVPR 2025](#)], a framework for spatiotemporal localization of small and fleeting objects in long videos
- Demonstrated precise localization in real-world conditions, achieving 49% relative improvement over prior SotA on Ego4D

Allen Institute for AI

Research Intern

Seattle, WA

Oct 2022 - Dec 2023

- Contributed to Unified-IO 2 [[CVPR 2024](#)], the first autoregressive multimodal model that can understand and generate images, text, audio, and actions

National University of Singapore

Research Assistant (Advisor: Prof. Kenji Kawaguchi and Prof. Alex Lamb)

Remote

Apr 2022 - Aug 2022

- Designed (1) an active learning algorithm robust to heteroskedastic noise; and (2) a self-supervised fine-tuning method to leverage unlabeled data in active learning setting, improving performance by 15% and reducing annotation costs [[ECAI 2023](#)]

Mila - Quebec AI Institute

Research Intern (Advisor: Prof. Yoshua Bengio)

Montreal, QC

Apr 2021 - Nov 2021

- Exposed limitations of uncertainty-based active learning by introducing three heteroskedastic data distributions [[NeurIPS 2021](#)]
- Developed adversarial training that reduces clean-data error by 48% while preserving robustness to adversarial attacks [[INNS 2022](#)]

- Leveraged image-based representation of malware binaries and ensembling to develop state-of-the-art model for malware detection and classification [[MMTA 2022](#), [CONIT 2021](#)]

SELECTED PUBLICATIONS & PREPRINTS

A full list of publications can be seen on my [Google Scholar](#) author page.

(* denotes equal contribution, † denotes alphabetical order)

1. **T-REN: Learning Text-Aligned Region Tokens Improves Dense Vision-Language Alignment and Scalability** [Link](#)
Savya Khosla, Sethuraman TV, Aryan Chadha, Alex Schwing, Derek Hoiem
European Conference on Computer Vision, 2026
2. **ViTAL-X: Video-Text Alignment with Cross-Modal Temporal Edits**
Sethuraman TV, Savya Khosla, Onkar Kishor Susladkar, Aditi Tiwari, Seoung Oh, Kushal Kafle, Joon-Young Lee, Derek Hoiem
European Conference on Computer Vision, 2026
3. **Stress Tests REVEAL Fragile Temporal and Visual Grounding in Video-Language Models** [Link](#)
Sethuraman TV, Savya Khosla, Aditi Tiwari, Vidya Ganesh, Rakshana Jayaprakash, Aditya Jain, Vignesh Srinivasakumar, Onkar Susladkar, Srinidhi Sunkara, Aditya Shanmugham, Rakesh Vaideeswaran, Abbaas Nishar, Simon Jenni, Derek Hoiem
Under review, 2026
4. **REN: Fast and Efficient Region Encodings from Patch-Based Image Encoders** [Link](#)
Savya Khosla, Sethuraman TV, Barnett Lee, Alex Schwing, Derek Hoiem
Neural Information Processing Systems, 2025
5. **RELOCATE: A Simple Training-Free Baseline for Visual Query Localization Using Region-Based Representations** [Link](#)
Savya Khosla, Sethuraman TV, Alex Schwing, Derek Hoiem
Computer Vision and Pattern Recognition, 2025
6. **MAGNET: Augmenting Generative Decoders with Representation Learning and Infilling Capabilities** [Link](#)
Savya Khosla, Aditi Tiwari, Kushal Kafle, Simon Jenni, Handong Zhao, John Collomosse, Jing Shi
Association for Computational Linguistics, 2025
7. **FRAME: Feature Representation and Anticipation with Memory** [Link](#)
Sethuraman TV, Savya Khosla, Vignesh Srinivasakumar, Jiahui Huang, Seoung Oh, Simon Jenni, Derek Hoiem, Joon-Young Lee
ArXiv, 2025
8. **Unified-IO 2: Scaling Autoregressive Multimodal Models with Vision, Language, Audio, and Action** [Link](#)
Jiasen Lu*, Christopher Clark*, Sangho Lee*, Zichen Zhang*, Savya Khosla, Ryan Marten, Derek Hoiem, Aniruddha Kembhavi
Computer Vision and Pattern Recognition, 2024
9. **Understanding and Improving Neural Active Learning on Heteroskedastic Distributions** [Link](#)
Savya Khosla, Chew Kin Whye, Jordan T. Ash, Cyril Zhang, Kenji Kawaguchi, Alex Lamb
European Conference on Artificial Intelligence, 2023
10. **Interpolated Adversarial Training: Achieving Robust Neural Networks Without Sacrificing Too Much Accuracy** [Link](#)
Alex Lamb, Vikas Verma, Kenji Kawaguchi, Alexander Matyasko, Savya Khosla, Juho Kannala, Yoshua Bengio
Neural Networks, 2022
11. **S-DCNN: Stacked Deep Convolutional Neural Networks for Malware Classification** [Link](#)
Anil Singh Parihar, Shashank Kumar, Savya Khosla
Multimedia Tools and Applications, 2022
12. **AE-DCNN: Autoencoder Enhanced Deep Convolutional Neural Network For Malware Classification** [Link](#)
Shashank Kumar*, Savya Khosla*, Shivangi Meena, Anil Singh Parihar
International Conference on Intelligent Technologies, 2021
13. **MuRIL: Multilingual Representations for Indian Languages** [Link](#)
Simran Khanuja, Diksha Bansal†, Sarvesh Mehtani†, Savya Khosla†, Atreyee Dey, Balaji Gopalan, Dilip Kumar Margam, Pooja Aggarwal, Rajiv Teja Nagipogu, Shachi Dave, Shruti Gupta, Subhash Chandra Bose Gali, Vish Subramanian, Partha Talukdar
ArXiv, 2021
Media Coverage: [Economic Times](#), [Indian Express](#), [Google AI Blog](#)

INDUSTRY EXPERIENCE

Google

Software Engineer

Bangalore, KA

Aug 2021 - Mar 2022

- Improved Google Search's web ranking infrastructure using deep learning for better multimodal document understanding
- Enhanced precision and recall in salient entity extraction from webpages by transitioning from traditional ML methods to LLMs

Google

Software Engineering Intern

Bangalore, KA

May 2020 - Jul 2020

- Initiated the development of MuRIL, a BERT-based multilingual language model for 17 Indian dialects
- Achieved 10.42% F1 improvement in sentiment analysis and 9.87% in named entity recognition for Indian languages

TEACHING EXPERIENCE

CS 445: Computational Photography

Teaching Assistant (Instructor: Prof. Derek Hoiem)

Urbana, IL

Aug 2023 - Dec 2023

CS 225: Data Structures and Algorithms with C++

Teaching Assistant (Instructor: Prof. Carl Evans and Prof. Brad Solomon)

Urbana, IL

Aug 2022 - May 2023

COURSES & CERTIFICATIONS

Graduate Courses

- CS 598: Machine Learning Algorithms for LLMs by Prof. Tong Zhang
- CS 598: Vision by Prof. Svetlana Lazebnik
- CS 588: Autonomous Vehicle System Engineering by Prof. David Forsyth
- CS 543: Computer Vision by Prof. Svetlana Lazebnik
- CS 445: Computational Photography by Prof. Derek Hoiem
- CS 444: Deep Learning for Computer Vision by Prof. Svetlana Lazebnik
- CS 410: Text Information System by Prof. ChengXiang Zhai
- CS 491: Competitive Programming by Prof. Mattox Beckman

Undergraduate Courses

- CO 201: Data Structures
- CO 202: Database Management System
- CO 203: Object-Oriented Programming
- CO 206: Algorithm Design and Analysis
- CO 304: Artificial Intelligence
- CO 404: Data Warehousing and Data Mining
- CO 407: Distributed Systems
- CO 414: Big Data Analytics
- CO 423: Swarm and Evolutionary Computing
- IT 420: Computer Vision

Online Courses & Certifications

- Deep Learning Specialization by Andrew Ng
- CS229 Lectures by Andrew Ng
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning by deeplearning.ai
- Competitive Programming Bootcamp by Coding Blocks
- Machine Learning Master Course by Coding Blocks

PROFESSIONAL SERVICES

Organizer: [UIUC Vision Lunch](#) (Aug 2025 - Present), [Vision Mini-Conference](#) (2025)

Reviewer: CVPR, ECCV, NeurIPS, TMLR

Mentor: [Undergraduate Research in Scientific Advancement](#) (Fall 2025)